SUB: Machine Learning (ML)
Date: 20 Sep 2026
Mid Sem Regular

Question

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An optimization-based machine learning model is being trained to minimize a loss function. At a particular stage of training, the model has a current parameter value of (w=4). The gradient of the loss function with respect to this parameter is $\partial L/\partial w = 12$, and the optimization algorithm initially uses a learning rate $\eta = 0.1$.

a) Determine the parameter value (w) obtained after one optimization update using the gradient-descent principle. Show the calculation clearly. **[1 Mark]**

b) The training configuration is subsequently modified by increasing the learning rate to $\eta = 0.5$, while keeping the current parameter value and gradient unchanged. Determine the resulting parameter value. Compare the two parameter updates quantitatively and explain how the choice of learning rate can influence the stability and convergence of the optimization process. **[1 Mark]**

c) During another training run, the following loss values are recorded:

Epoch	Loss
1	2.8
2	4.1
3	7.6
4	15.3

Examine the observed loss progression and infer the likely behaviour of the optimization algorithm. Identify the most plausible underlying issue and justify your conclusion based on the trend in the loss values. **[2 Marks]**

d) Suppose you are responsible for modifying the training configuration to address the observed instability. Propose **two appropriate modifications** to the optimization procedure. For each modification, explain how it would alter the parameter-update process and why it could help the model move toward a stable minimum rather than diverging. **[2 Marks]**

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A linear regression model is being developed to estimate house prices using two input attributes:

x_1: house area in square feet, ranging from 600 to 4500
x_2: number of bedrooms, ranging from 1 to 5

During model training using gradient descent, the optimization process exhibits the following behaviour: the cost function decreases very slowly, and successive parameter updates appear to move repeatedly from one side of the minimum to the other rather than approaching it directly.

a) Based on the characteristics of the input attributes and the observed optimization behaviour, infer the most likely factor responsible for the inefficient convergence. Explain your reasoning. **[1 Mark]**

b) Explain how a substantial difference in the numerical scales of two input attributes can cause the corresponding parameter updates to have significantly different magnitudes. Relate your explanation to the gradient-descent update rule. **[2 Marks]**

c) The house-area attribute has a mean of (2200) ft² and a standard deviation of (800) ft². For a house with an area of (3000) ft², determine its standardized value using an appropriate technique and interpret the resulting transformed value. **[1 Mark]**

d) After applying the transformation in part (c) to the relevant feature(s), predict how the geometry of the cost function would change. In particular, discuss the expected effect on the **shape of the cost contours** and the **trajectory followed by gradient descent**. Justify why the transformation can improve convergence without changing the underlying relationship being learned. **[2 Marks]**

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Consider the following dataset of 4 students, where each instance includes:

x_1: Number of hours studied

x_2: Number of hours slept

y: Outcome (1 if the student passed the exam, 0 if the student failed)

Student	Hours Studied (x_1)	Hours Slept (x_2)	Outcome (y)
1	4	6	1
2	2	4	0
3	5	5	1
4	1	3	0

Let $\vec{w} = [0.6, 0.4] and\ b = -2$

Based on the above dataset, answer the following questions:

(a) For Student 1 and Student 2 only, calculate $z=w^{T}x+b$, the sigmoid probability, and the predicted class using threshold 0.5. **[2 Marks]**

(b) Student 1 and Student 3 both belong to the positive class. Compare their predicted probabilities and explain what the difference tells you about the model's confidence. **[1 Mark]**

(c) Suppose the classification threshold is increased from 0.5 to 0.7. Without retraining the model, explain how this could affect the predicted classes and the trade-off between false positives and false negatives. **[2 Marks]**

(d) If reducing false negatives is more important than reducing false positives in this application, would you prefer a higher or lower classification threshold? Justify. **[1 Mark]**

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A university wants to develop a simple classification model to identify whether a student is likely to **Pass** or **Fail** based on attendance, study behaviour, and assignment submission. The following observations have been collected:

Student	Attendance	Study Hours	Assignment	Result
S1	High	High	Submitted	Pass
S2	High	Low	Submitted	Pass
S3	Low	High	Submitted	Pass
S4	Low	Low	Not Submitted	Fail
S5	High	Low	Not Submitted	Fail
S6	Low	Low	Submitted	Fail
S7	High	High	Submitted	Pass
S8	Low	High	Not Submitted	Fail

The university decides to use the **ID3 decision-tree algorithm**, selecting each splitting attribute based on the highest information gain.

a. Before selecting the first splitting attribute, determine the uncertainty associated with the target variable **Result** by calculating its entropy. **[1 Mark]**
b. Apply the ID3 criterion systematically to the given observations. Compute the required information gain values at each stage and construct the **complete decision tree**, showing the attribute selected at each internal node and the corresponding leaf classifications. **[4 Marks]**
c. The model is subsequently presented with a new student whose characteristics are:

- **Attendance:** Low
- **Study Hours:** High
- **Assignment:** Not Submitted

Use the decision tree obtained in part (b) to determine the predicted outcome. Clearly indicate the sequence of decisions (decision path) used to arrive at the final classification.

[1 Mark]

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a) A company is building a Logistic Regression model to estimate the likelihood that a customer will purchase a product. After training, the model assigns the following coefficients to five input attributes:

Feature	Weight
Age	2.0
Income	0.8
Website Visits	0.05
Previous Purchases	1.5
Social Media Activity	0.02

The data scientist is concerned that the model may be unnecessarily complex because some attributes contribute very little through their learned coefficients. A regularization parameter ($\lambda = 0.5$) is introduced.

i. Quantify the additional penalty imposed on the model under **L1** and **L2** regularization using the given coefficients and λ. Show the necessary calculations. **[2 Marks]**

ii. Suppose the objective is to obtain a simpler model in which features making negligible contributions can be effectively excluded. Between L1 and L2 regularization, determine which approach would better support this objective. Justify your choice based on how each penalty influences model coefficients. **[1 Mark]**

iii. Consider an alternative situation in which retaining the contribution of every feature is important, but the data scientist wants to discourage excessively large coefficients. Select the more appropriate regularization approach and justify your decision in terms of its effect on the learned weights. **[1 Mark]**

b. A team member proposes using the **Mean Squared Error (MSE)** as the primary cost function for training this Logistic Regression model. Evaluate this proposal and explain the fundamental reason why MSE is generally unsuitable as the optimization objective for Logistic Regression. Your answer should relate the issue to the model's output function and the resulting optimization behaviour. **[2 Marks]**